

WASP-104b

R-2458

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_180423 · created 2026-08-23T00:36:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458231.7315 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.035
Transit depth (Rp/Rs)^2	2.01 +/- 1.0 [%]
Orbital Inclination (inc)	81.8 +/- 1.93
Airmass coefficient 1 (a1)	1.078 +/- 0.016
Airmass coefficient 2 (a2)	-0.062 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.44 %
Best Comparison Star	#3 - [341, 289]
Optimal Aperture	3.2
Optimal Annulus	4.44
Transit Duration (day)	0.05 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.035 vs 0.1214 (deviation 0.51σ)
Duration fit vs published	0.05 d vs 0.0742 d (deviation 0.72σ)
Mid-time O–C	-3.4 min
Depth SNR	3.5
Residual scatter	1.44 %

Light curve

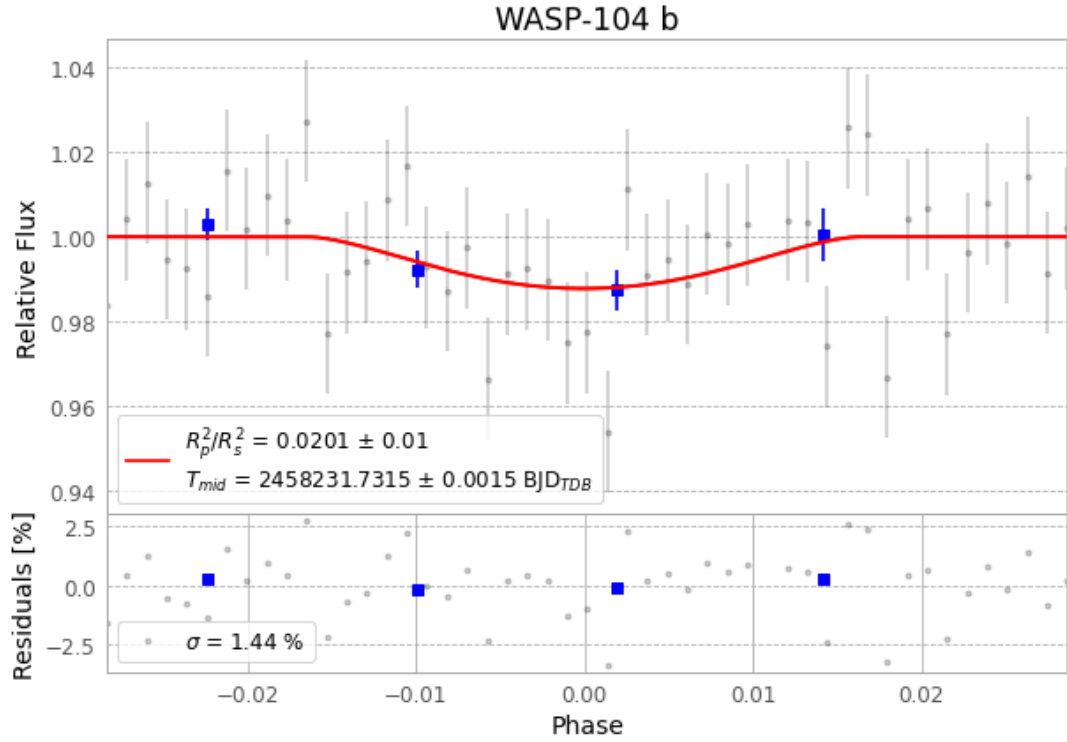


Figure 1 — FinalLightCurve_WASP-104 b_2018-04-22.png (SHA-256 90271a654ae7f2b4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [327, 246], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 2149277b8174f783...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

49 FITS frames (32 MB), WASP-104180423041512.FITS → WASP-104180423063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-180423.log (79df968069c9...), FinalLightCurve_WASP-104 b_2018-04-22.png (90271a654ae7...), FinalLightCurve_WASP-104 b_2018-04-22.csv (dde74b61f5b4...)

Compute resources

Wall time 174.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 00:36Z.